

Hardware-Aware Quantum Bayesian Learner Compression: Linking Human Belief Updating to NISQ Circuit Complexity

Mohammad Zoraiz

Pratt School of Engineering, Duke University

Electrical Engineering, Physics, and Computer Science, Durham, NC, USA

mz248@duke.edu

Abstract—We connect quantum models of cognition with the constraints of near-term quantum hardware. A small *quantum Bayesian learner*—a belief state on a two-qubit register, a prior, and a stimulus-dependent evidence channel—learns a similarity-structured categorization task, with the posterior category read out by measurement. Making the learner a hardware-efficient variational circuit, we add a hardware-aware penalty equal to the real post-transpile two-qubit gate count on IBM’s *FakeManila* backend and compress it by greedy structured pruning of individual Heisenberg interaction terms. Across three difficulty levels and five seeds, the accuracy–complexity frontier reveals a sharp *capacity boundary*: structured compression is essentially free—and on the hardest task beneficial—yet removing all entanglers collapses the learner to chance. Learned masks beat random masks at every matched budget. On a physical IBM device (*ibm_fez*), the full circuit collapses to near chance while the compressed circuit stays accurate: on real NISQ hardware, hardware-aware compression is not merely economical but necessary.

Index Terms—Quantum cognition, Bayesian learning, variational quantum circuits, hardware-efficient ansatz, NISQ devices, circuit compression.

I. INTRODUCTION

Bayesian models cast human learning as *belief updating*: a prior over hypotheses revised by evidence [1]. *Quantum cognition* represents belief states in Hilbert space—as density operators—and updates them with the mathematics of quantum channels, capturing interference and order effects [2]. In parallel, noisy intermediate-scale quantum (NISQ) devices [6] have made variational quantum algorithms the dominant near-term paradigm [5], built from hardware-efficient ansätze of single-qubit rotations and native two-qubit entanglers [3], [4]. Because two-qubit gates dominate the NISQ error budget [6], [7], reducing their number without destroying task performance is a central engineering objective.

We join these threads. The belief is a state on a two-qubit register, the evidence update is a stimulus-conditioned channel, and training is on a similarity-structured categorization task [9]; the update circuit is a Heisenberg-type hardware-efficient ansatz, and a hardware-aware loss penalizes the two-qubit gates remaining after transpilation. The guiding question is

concrete: *how much of the circuit can be removed before the learner can no longer represent the task?* Our contributions are (1) a compact quantum Bayesian learner in the density-operator/channel language, exactly simulated and analytically differentiable; (2) a hardware-aware cost equal to the real post-transpile two-qubit count on *FakeManila*, resolved per $R_{XX}/R_{YY}/R_{ZZ}$ term; (3) accuracy–complexity frontiers locating a *capacity boundary*; (4) a learned-versus-random ablation; and (5) validation on a physical IBM device.

II. METHODS

Belief states and evidence channels. The belief about a stimulus is a state ρ on $n=2$ qubits. A prior ρ_0 is updated by a completely-positive trace-preserving channel conditioned on the stimulus x , $\mathcal{E}_x(\rho) = \sum_k E_{x,k} \rho E_{x,k}^\dagger$, with the posterior read out by measurement [2]. A fast *pure-state* model takes $\rho_0 = |0\rangle\langle 0|^{\otimes n}$ with a unitary evidence channel; a literal *mixed-state* model starts from the maximally mixed prior $\mathbb{I}/2^n$ and applies a non-unital evidence channel—a stimulus-dependent Lüders/POVM filter with Bayesian trace renormalization plus amplitude damping—so the belief is a bona fide mixed state (verified PSD, unit-trace, purity < 1). Both are exactly simulated and analytically differentiable, and the mixed model reduces bit-for-bit to the pure one when the channel is off.

Ansatz and hardware-aware cost. Each of $D=4$ layers applies single-qubit R_X, R_Z rotations and a Heisenberg entangler $U_{ij} = \exp[-i(\alpha X_i X_j + \beta Y_i Y_j + \gamma Z_i Z_j)]$ realized as native R_{XX}, R_{YY}, R_{ZZ} , interleaved with data re-uploading [3], [4]. A per-term binary mask $m_{\ell,p} \in \{0, 1\}$ selects each interaction term (12 maskable terms; $R_{PP}(0) = \mathbb{I}$ when off), a small instance of quantum architecture search [8]. The cost $N_{2q}(m)$ is the real number of two-qubit gates surviving transpilation to *FakeManila* (including SWAPs), computed via Qiskit’s transpiler [10]; the full circuit gives $N_{2q}=24$. The objective is $L(\theta, m) = \frac{1}{N} \sum_k \text{BCE}(y_k, \hat{y}_k) + \lambda N_{2q}(m)$. The readout uses only *single-qubit* Pauli expectations ($\langle Z_0 \rangle, \langle Z_1 \rangle, \langle X_0 \rangle, \langle X_1 \rangle$) through a trainable head $\hat{y} = \sigma(w \cdot v + b)$; this makes entanglement the *only* carrier of feature interaction, so N_{2q} meaningfully bounds the learner’s capacity.

TABLE I
LEARNED VS. RANDOM MASKS (HARD TASK, MEAN TEST ACCURACY,
FIVE SEEDS). Δ IS THE LEARNED ADVANTAGE AT THE SAME TWO-QUBIT
BUDGET.

N_{2q}	Learned	Random	Δ
2	0.847	0.742	+0.104
6	0.927	0.814	+0.112
10	0.913	0.794	+0.119
14	0.917	0.826	+0.091
18	0.900	0.809	+0.091
22	0.897	0.836	+0.061

Compression and task. Parameters are trained by Adam with exact JAX autodiff through the belief-state simulator (verified bit-for-bit against Qiskit). *Greedy structured pruning* repeatedly removes the term whose removal least increases training cross-entropy, warm-start retraining after each step, tracing the accuracy- N_{2q} frontier from 24 to 0. The task is a checker-board categorization $y = (\lfloor fx_0 \rfloor + \lfloor fx_1 \rfloor) \bmod 2$ on $[0, 1]^2$ [9], with a difficulty knob $f=2, 3, 4$ (easy/medium/hard); 200 balanced stimuli are split 70/30 (chance 0.5).

III. RESULTS

Capacity boundary. The learner is well above chance at all nonzero budgets (easy/medium ≈ 0.95 – 0.97 , hard ≈ 0.87 – 0.93) and the frontier is nearly flat, so most entangling structure is redundant; at $N_{2q}=0$ every difficulty collapses toward chance (0.52/0.64/0.49). Some entanglement is thus necessary but far less than the full circuit. On the hard task accuracy *peaks* at intermediate cost (0.927 at $N_{2q}=6$) and is lower for the full circuit (0.873 at $N_{2q}=24$): moderate compression acts as a capacity-control regularizer.

Structure beats count. At matched budgets, greedy (learned) masks beat random masks of identical sparsity at *every* budget by ~ 6 – 12 points (Table I); per seed, the learned mask wins 49 of 55 paired comparisons, mean advantage $+0.083$, one-sided Wilcoxon $p < 10^{-7}$. *Which* entanglers are kept matters, not only how many.

Validation on hardware. On a physical IBM processor (ibm_fez, a 156-qubit Heron device, hard task), the on-device frontier *reverses* the simulated picture (Fig. 1): accuracy falls monotonically with N_{2q} —0.81 at $N_{2q}=2, 6$, 0.75 at 12, and 0.56 (near chance) at 18, 24. A denser endpoint run confirms the full circuit at 0.60 (simulation 0.95) versus the compressed circuit at 0.83 (simulation 0.93). On real NISQ hardware, surplus entangling capacity is actively harmful, and task-aware pruning is what makes the learner usable.

Robustness. Under the FakeManila noise model (4096 shots, five seeds) the noisy frontier tracks the noiseless one within a few percent at every budget. The mixed-state model reproduces all findings (hard peak 0.88 at $N_{2q}=6$ vs 0.87 at 24; learned>random, mean $+0.043$). Classical baselines on identical splits confirm the difficulty axis: logistic regression stays at chance, while RBF-SVM and MLP fall monotonically from easy to hard.

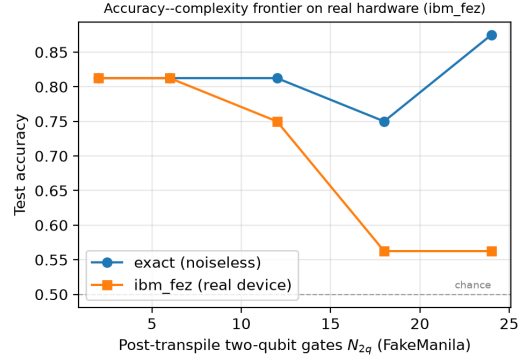


Fig. 1. Accuracy-complexity frontier on a physical IBM device (ibm_fez, hard task). In simulation accuracy is flat; on hardware it falls monotonically with N_{2q} , collapsing toward chance for the full circuit. The widening gap is the cost of redundant entangling capacity—and the case for compression.

IV. CONCLUSION

Compressing a quantum Bayesian learner by greedy structured pruning reveals a clear capacity boundary: most entangling structure is redundant—the learner sheds roughly two-thirds of its two-qubit gates for free, and on the hardest task a compressed circuit beats the full one—yet entanglement remains necessary, since removing it collapses the learner to chance. Structure beats count, with learned masks outperforming random masks of equal budget. These conclusions are not artifacts of simulation: on a physical IBM device the full circuit collapses to near chance while the compressed circuit stays accurate. The result links cognitive capacity and quantum-circuit capacity in one experiment, and shows that quantum models of cognition can benefit from—not merely tolerate—the constraints of present-day hardware. Code and logs are public.¹

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¹<https://github.com/zoraizmohammad/qb-learner-compression>